ISEN 637 Lecture 11

Advantage Estimation and Advantage Actor-Critic

Outline

1. Policy Gradient with a Baseline
2. Why $V_\theta(s)$ is a Good Baseline ?
3. Advantage Estimation
4. Advantage Actor-Critic Algorithm

Policy Gradient with a Baseline

- Recall that for infinite horizon:

$$\nabla V_\theta(s') = \mathbb{E}_{s\sim d_\theta(s'),a\sim\pi_\theta}\left[\nabla \log \pi_\theta(a|s) Q_\theta(s,a)\right]$$

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- An estimate of the form:

$$\hat{Q}_\theta(s_t, a_t) = \sum_{n=t}^{\infty} \gamma^{n-t} R(s_n, a_n)$$

  has high variance.

- How to reduce variance without introducing bias ?

Baselines Do Not Change the Policy Gradient

- Consider an estimate $\hat{Q}_\theta(s_t, a_t) - b(s)$ where the baseline $b(s)$ independent of a. Note that:

$$\begin{aligned}
\mathbb{E}_{a\sim\pi_\theta}\left[\nabla \log \pi_\theta(a|s) b(s)\right] &= b(s) \sum_a \pi_\theta(a|s) \nabla \log \pi_\theta(a|s) \\
&= b(s) \sum_a \nabla \pi_\theta(a|s) \\
&= b(s) \nabla \big(\sum_a \pi_\theta(a|s) \big) = b(s) \nabla 1 = 0
\end{aligned}$$

- Thus, any state-dependent baseline preserves unbiasedness:

$$\begin{aligned}
\mathbb{E}_{s\sim d_\theta(s'), a\sim\pi_\theta}\left[\nabla \log \pi_\theta(a|s)(Q_\theta(s,a) - b(s))\right] &= \nabla V_\theta(s') \\
&\quad - \mathbb{E}_s \mathbb{E}_{a\sim\pi_\theta}\left[\nabla \log \pi_\theta(a|s) b(s)\right] \\
&= \nabla V_\theta(s')
\end{aligned}$$

Why $V_\theta(s)$ is a Good Baseline ?

- Let us set the value function s baseline:

$$b(s) := V_\theta(s) = \sum_a \pi_\theta(a|s) Q_\theta(s,a)$$

and define the advantage of action a in state s under policy π_θ as:

$$A_\theta(s,a) := Q_\theta(s,a) - V_\theta(s)$$

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- Note that:
  - If $A_\theta(s,a) > 0$ implementing action a when in state s and then following π_θ yields performance improvement.
  - If $A_\theta(s,a) < 0$ implementing action a when in state s and then following π_θ yields performance deterioration.
  - On average:

$$\begin{aligned}\mathbb{E}_{a\sim\pi_\theta}[A_\theta(s,a)] &= \mathbb{E}_{a\sim\pi_\theta}[Q_\theta(s,a) - V_\theta(s)] \\ &= \mathbb{E}_{a\sim\pi_\theta}[Q_\theta(s,a)] - V_\theta(s) = 0\end{aligned}$$

Why is $V_\theta(s)$ a Good Baseline?

- For fixed $s \in \mathcal{S}$, define

$$X := \nabla \log \pi_\theta(a|s) \left(Q_\theta(s,a) - b(s)\right).$$

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- For fixed $s \in \mathcal{S}$, define

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- We minimize a scalar measure of variance:

$$\text{Trace}(\text{Cov}[X]) = \mathbb{E}[X^\top X] - \mathbb{E}[X]^\top \mathbb{E}[X]$$

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- Since

$$\mathbb{E}[X] = \mathbb{E}\left[\nabla \log \pi - \theta(a|s) Q - \theta(s,a)\right]$$

does not depend on $b(s)$:

minimizing $\mathbb{E}[X^\top X]$ *is equivalent to minimizing* $\textit{Trace}(\textit{Cov}[X])$

Why is $V_\theta(s)$ a Good Baseline?

- Note that

$$\mathbb{E}[X^\top X] = \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\left[\|\nabla \log \pi_\theta(a|s)\|^2 (Q_\theta(s,a) - b(s))^2\right]$$

- First order condition is:

$$\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\left[\|\nabla \log \pi_\theta(a|s)\|^2 (Q_\theta(s,a) - b(s))\right] = 0.$$

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- Hence the optimal baseline is:

$$b^*(s) = \frac{\mathbb{E}_{a\sim\pi_\theta}\left[\|\nabla \log \pi_\theta(a|s)\|^2 Q_\theta(s,a)\right]}{\mathbb{E}_{a\sim\pi_\theta}\left[\|\nabla \log \pi_\theta(a|s)\|^2\right]}.$$

Why is $V_\theta(s)$ a Good Baseline?

- If gradient magnitudes are similar across actions,

$$b^*(s) \approx \mathbb{E}_{a\sim\pi_\theta}[Q_\theta(s,a)] = V_\theta(s).$$

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- That is $V_\theta(s)$ is near-optimal baseline.

Advantage and First-Order Optimality

- Recall the policy gradient:

$$\nabla V_\theta(s') = \mathbb{E}_{s\sim d_\theta(s'),\, a\sim\pi_\theta(\cdot|s)}\left[\nabla \log \pi_\theta(a|s)\, A_\theta(s,a)\right].$$

- At an interior optimum θ^*:

$$\nabla V_{\theta^*}(s') = 0.$$

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- At an interior optimum θ^*:

$$\nabla V_{\theta^*}(s') = 0.$$

- In the neighborhood of θ^*:

$$A_{\theta^*}(s,a) \leq 0 \quad \text{for all } a \in \mathcal{A}$$

and

$$A_{\theta^*}(s,a) = 0$$

for actions a such that $\pi^*_\theta(a|s) > 0$.

Monte Carlo Advantage Estimator

- Recall advantage:

$$A_\theta(s_t, a_t) = Q_\theta(s_t, a_t) - V_\theta(s_t)$$

- A Monte Carlo estimate:

$$\hat{A}_t^{MC} = \left(\sum_{k=t}^{\infty} \gamma^{k-t} R_k \right) - \hat{V}_\phi(s_t)$$

- However, this estimator is unbiased (if episodes complete) and very high variance

TD Residual as Advantage Estimator

- Recall the TD residual:

$$\delta_t = R_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

- If $V_\phi = V_\theta$ then:

$$\mathbb{E}[\delta_t | s_t, a_t] = Q_\theta(s_t, a_t) - V_\theta(s_t) = A_\theta(s_t, a_t)$$

That is, TD is an unbiased one-step advantage estimator.

Generalized Advantage Estimation (GAE)

- Recall the TD residual:

$$\delta_t = R_t + \gamma V(s_{t+1}) - V(s_t)$$

and define GAE(λ):

$$\hat{A}_t^{GAE(\lambda)} := \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}$$

- With $\lambda = 0$ this is the TD advantage.
- With $\lambda = 1$ this is the MC advantage.

Advantage Actor–Critic Algorithm

- Critic update:

$$\phi \leftarrow \arg\min_{\phi} \mathbb{E}[\delta_t^2]$$

- Actor update:

$$\theta \leftarrow \theta + \alpha \mathbb{E}\left[\nabla \log \pi_\theta(a_t|s_t)\hat{A}_t\right]$$